Land cover / land use classifier: test on 500 records

Run 20261012-151353 · 500 random records (seed 42) of 25,129 · both taxonomies, depth 3 · mistral-small-latest, 5 records/call, 8 parallel · 12 Oct 2026

11 min 29 s
total time (0.73 records/s)

$0.183
total cost ($0.37 per 1,000)

462
API calls

936k
tokens (about 1,870 per record)

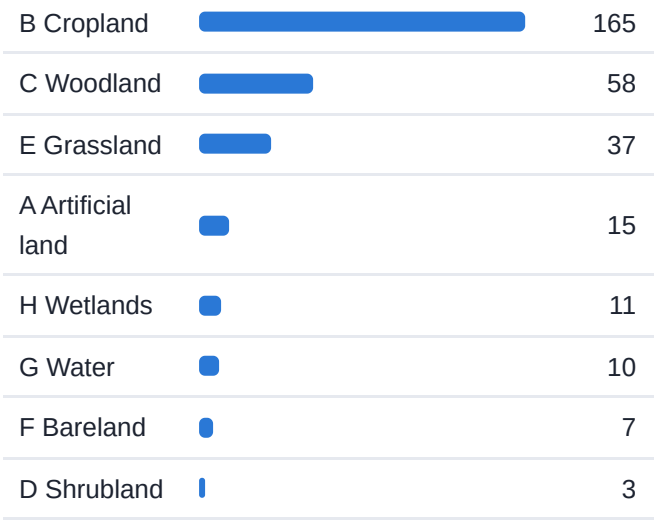
Time and cost

METRIC	VALUE
Wall time	689 s (1.38 s per record)
Input / output tokens	841,323 / 94,913
Cost per record	$0.00037 (prices: $0.15 / $0.60 per million input / output tokens)
Extrapolated to all 25,129 records	about 9.5 h, about $9.20

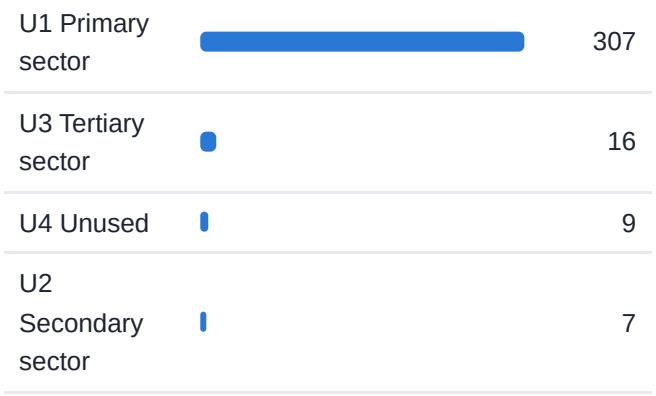
Label coverage and confidence (default thresholds)

	LAND COVER	LAND USE
Records with at least 1 label	284 (57%)	337 (67%)
Records reaching level 2 / level 3	37% / 20%	52% / 42%
Labels per record (deepest)	0.71	0.71
Mean level-1 confidence	0.94	0.92
Level-1 labels with confidence ≥ 0.8	100%	99.7%

Land cover: level 1 classes



Land use: level 1 classes



Accuracy: agreement with a reference run

The records table has no human labels, so true accuracy cannot be computed. This is the agreement with run 20261007-162217 (1 record per call) on the 100 records both runs contain. The reference is not human-verified: the table measures consistency, not correctness. Precision, recall and F1 treat the reference labels as truth.

LEVEL	LAND COVER				LAND USE			
	PREC.	RECALL	F1	EXACT	PREC.	RECALL	F1	EXACT
1	88%	85%	0.86	85%	94%	89%	0.92	90%
2	78%	78%	0.78	83%	85%	76%	0.80	80%
3	91%	83%	0.87	94%	82%	71%	0.76	79%

Reading the results. Level 1 agrees well (F1 0.86 land cover, 0.92 land use). Land use level 3 is the weakest (F1 0.76, 79% exact match). Land use is dominated by U1 (307 of 337 labelled records). A record gets no label when no class passes its threshold; this was not checked against the abstracts. For real accuracy, hand-label about 100 records and compare.